JIANGZI CHEN

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EDUCATION

Arizona State University (ASU) 8/2025–Expected 5/2027
M.S. in Civil, Environmental and Sustainable Engineering
GPA: 3.5/4.0

Nanjing University of Information Science and Technology (NUIST) 9/2021–6/2025
B.S. in Physical Geography and Resource Environment
GPA: 3.46/4 (83)

PUBLICATIONS

Wan, Z., Zhao, C., Zhu, J., Ma, X., **Chen, J.**, & Wang, J. (2024). Assessment and prediction of coastal ecological resilience based on the Pressure–State–Response (PSR) model. *Land*, 13(12), 2130. https://doi.org/10.3390/land13122130

Impact of *Spartina alterniflora* expansion on tidal creek formation in the mudflat of the Yellow Sea. Manuscript submitted.

CONFERENCE PRESENTATIONS

Chen, J., & Zeng, R. (2026). Estimation of Unpermitted Wastewater Treatment Facility (OWTF) in Rural Arizona for Environmental Health. Accepted poster presentation, Session H43P: Water Quality and Watersheds: From Scientific Innovations to Actions II. AGU Fall Meeting 2026, San Francisco, CA, December 10, 2026.

RESEARCH EXPERIENCE

Fine-Scale Hydrologic Feature Detection from High-Resolution LiDAR-Derived DEMs 10/2026–Present
Supervisor: Dr. Ruijie Zeng, ASU

- Redesigning the experiments of a prior doctoral study in the group that uses partial differential equation (PDE)-based anisotropic diffusion smoothing of sub-meter LiDAR-derived DEMs to detect sinkholes and drainage networks
- Identifying a threshold for the number of smoothing iterations, so that the method removes noise while preserving edges without case-by-case tuning and generalizes across datasets and landscapes

Agentic AI System for Automating Hydrological and Atmospheric Modeling Workflows 9/2026–Present
Independent project (conceived and led); supported by Dr. Ruijie Zeng, ASU

- Designed and developed an agentic AI system that automates routine modeling tasks, including model configuration, execution, and input/output format alignment across models
- Currently runs the hydrological and land surface models SWAT and VIC; gained hands-on experience with WRF and Noah-MP workflows and data structures during development

Unpermitted On-site Wastewater Treatment Facility (OWTF) Identification in Rural Arizona 1/2026–Present
Supervisor: Dr. Ruijie Zeng, ASU; project with the Arizona Department of Environmental Quality (ADEQ)

- Conducted GIS-based spatial analysis of on-site wastewater treatment facility (OWTF) distribution across Arizona
- Developed a novel classification method for identifying potential unpermitted OWTFs
- Results accepted for poster presentation at the AGU Fall Meeting 2026

Coastal Ecological Resilience Assessment and Prediction Based on the PSR Model 2/2024–7/2025
Supervisor: Prof. Chengyi Zhao, NUIST

- Developed a Pressure-State-Response (PSR)-based approach to assess and predict pixel-scale, multi-year ecological resilience (ER)
- Analyzed spatiotemporal variations of ER in China's coastal zone over recent decades and projected future ER trends under multiple SSP-RCP scenarios
- Co-authored the resulting publication in *Land* (2024)

Calculation of Ecological Water Use in the Tarim River Basin 6/2024–7/2025
Supervisor: Yaojie Liu, NUIST

- Extracted NDVI for the Tarim River Basin using Google Earth Engine (GEE)
- Removed outliers through interpolation and other methods, and cleaned and prepared data for further analysis

Impact of *Spartina alterniflora* Expansion on Tidal Creeks in the Yellow Sea Mudflat 10/2023–8/2024
Supervisor: Prof. Guanghui Zheng, NUIST

- Used high-resolution remote sensing to monitor the invasion of *S. alterniflora* in the Yancheng Red-crowned Crane Reserve and its impact on tidal creek evolution
- Quantified relationships between the distribution and density of *S. alterniflora* and tidal creek characteristics, including bifurcation ratio (Rb), curvature, and density

Regional Disaster Resilience Analysis Based on a Multi-Attribute Decision-Making Model 9/2022–7/2024
Supervisor: Assoc. Prof. Jiaxun Li, NUIST

- Extracted geographic information for Jiangning District, Nanjing, using ArcGIS and ENVI
- Assessed regional resilience with a hybrid MADM model, identified factors driving differences in resilience, and proposed strategies to improve village resilience and reduce disaster losses

Evolution and Optimization of Coastal Zone Ecological Patterns Based on Cellular Automata 3/2023–3/2024
Project Leader; Supervisor: Prof. Chengyi Zhao, NUIST

- Proposed an integrated framework coupling the PLUS land-use model with the InVEST model
- Projected land-use change and quantified habitat quality in China's coastal zone from 2020 to 2050 under three SSP-RCP scenarios (SSP1-2.6, SSP2-4.5, and SSP5-8.5)
- Evaluated the driving forces of habitat quality change and the contribution of land-cover change to habitat degradation

TEACHING EXPERIENCE

Teaching Assistant, CEE 441, Arizona State University Spring 2026

- Held regular office hours to answer students' questions and graded assignments

PROFESSIONAL EXPERIENCE

Implementation Engineer Intern, GTMAP 7/2024–8/2024
Monitoring and Evaluation of the Rural Planning Initiative in Jiangsu Province

- Interpreted satellite imagery to identify site types, calculated area statistics, and determined potential development areas
- Drafted initial neighborhood plans, positioning enterprises, farmland, transportation lines, and residential areas for a rational layout

Nanjing Bureau of Planning and Natural Resources 1/2023–2/2023

- Inspected 22 geological disaster sites in Gulou District, Nanjing, assessed hazards and risks to nearby residents, and classified site danger levels
- Monitored land use for illegal or improper purposes and verified floor-space and planning records for new residential areas

LEADERSHIP

Member, Student Union, Changwang School of Honors, NUIST 9/2021–7/2022

- Handled external relations and coordinated communications for large-scale student events

SKILLS

GIS & Remote Sensing: ArcGIS, ENVI, Google Earth Engine (GEE), Fragstats; LiDAR-derived DEM analysis
Environmental Modeling: PLUS, InVEST, SWAT, VIC; hands-on experience with WRF and Noah-MP
Programming & AI: Python, MATLAB; agentic AI system development